

GRAVEATLAS

Phase 10 — Long-Term Expansion & Advanced GraveAtlas Ecosystem

GRAVEATLAS — PHASE 10
LONG-TERM EXPANSION & ADVANCED GRAVEATLAS ECOSYSTEM

PHASE 1 THROUGH PHASE 9 MUST ALREADY BE COMPLETE.

Continue from the existing GraveAtlas project.

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CORE RULES
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1. Do NOT rebuild Phases 1-9.
2. Reuse existing architecture, data, APIs, security, provenance, moderation, audit, release and operational systems.
3. Do NOT start Phase 11.
4. Do NOT add paid services without explicit approval.
5. Prefer existing/free infrastructure.
6. Do NOT expose secrets, credentials, tokens or private data.
7. Do NOT fabricate historical facts, records, users, analytics, coverage, performance or test results.
8. Do NOT silently alter authoritative historical data.
9. Preserve provenance and source attribution for every historical record.
10. Do NOT perform destructive bulk operations without explicit approval.
11. AI must assist research, never silently invent historical facts.
12. Complete Phase 10 only, then stop.

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PART 1 — PHASE 9 BASELINE
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Read the actual Phase 9 implementation and final report.

Identify:

- completed Phase 9 work
- remaining blockers
- measured limitations
- real user/data needs
- scalability constraints
- security constraints
- approved roadmap items

Do not assume Phase 9 was successful without evidence.

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PART 2 — ADVANCED CEMETERY DISCOVERY
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Improve discovery using actual data.

Support where justified:

- regional exploration
- cemetery collections
- historical collections
- geographic browsing
- related cemeteries
- source-based discovery

Do not manufacture relationships.

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PART 3 — ADVANCED HISTORICAL RESEARCH
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Provide research tools for connecting:

- people
- records
- cemeteries
- dates
- locations
- sources

Every relationship must have:

- evidence
- provenance
- confidence/uncertainty where appropriate

Never present an inferred relationship as confirmed fact.

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PART 4 — TIMELINE EXPLORATION

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Where sufficient evidence exists, allow users to explore:

- historical timelines
- cemetery history
- record chronology
- related events

Clearly distinguish:

CONFIRMED
INFERRED
UNKNOWN

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PART 5 – FAMILY & RECORD RELATIONSHIPS

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Where supported by reliable evidence, provide relationship exploration.

Never automatically create definitive family relationships from:

- similar names
- approximate dates
- location coincidence
- AI guesses alone

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PART 6 – ADVANCED MAP SYSTEM

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Evaluate support for:

- cemetery boundaries
- cemetery sections
- plot/area information
- historical locations
- historical map layers
- richer geographic metadata

Preserve the original source for every geographic claim.

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PART 7 – RESEARCH WORKSPACE

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Where justified, provide:

- saved research
- bookmarks
- collections
- research notes
- saved searches
- exportable research results

Private research information must be protected.

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PART 8 – DATA INTEROPERABILITY

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Design safe import/export capabilities.

Support standardized:

- records
- cemetery metadata
- locations
- sources
- provenance
- version information

Imports must pass:

VALIDATION
→ DUPLICATE CHECK
→ PROVENANCE CHECK
→ MODERATION
→ PUBLICATION

Never overwrite authoritative records automatically.

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PART 9 – GLOBAL EXPANSION

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Prepare architecture for:

- additional countries
- additional languages
- regional naming conventions
- local date formats
- local cemetery structures

Do not assume one country's cemetery model applies globally.

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PART 10 – INTERNATIONALIZATION
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Improve translation/localization architecture.

Preserve:

- original names
- original source text
- historical spellings
- transliterations where useful

Do not replace authoritative source names with machine translations.

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PART 11 – ADVANCED COMMUNITY RESEARCH
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Where justified, support:

- collaborative research
- research projects
- curated collections
- contributor collaboration

Maintain:

- permissions
- moderation
- audit history
- provenance

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PART 12 – LONG-TERM ARCHIVAL ARCHITECTURE
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Strengthen preservation of:

- dataset snapshots
- immutable versions
- source references
- provenance
- schema versions
- release history

Ensure historical changes remain traceable.

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PART 13 – AI RESEARCH ASSISTANCE
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AI may assist with:

- OCR assistance
- transcription suggestions
- duplicate detection
- record matching
- source comparison
- research assistance
- search assistance

AI MUST NOT:

- invent records
- invent sources
- fabricate people
- fabricate relationships
- silently change authoritative data
- present guesses as confirmed history

All AI-generated suggestions must be clearly identifiable and reviewable.

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PART 14 – AI CONFIDENCE & HUMAN REVIEW
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Where AI is used, maintain:

- input/source reference
- generated suggestion
- confidence or uncertainty
- human review status
- final decision
- audit history

Human approval is required before AI-generated historical information becomes authoritative.

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PART 15 – ADVANCED SEARCH
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Evaluate:

- faceted search
- date filtering
- geographic filtering
- source filtering
- cemetery filtering
- relationship filtering
- historical-period filtering

Maintain search correctness and performance.

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PART 16 – PUBLIC API / ECOSYSTEM

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If justified by Phase 9 evidence, design a public developer API.

Consider:

- authentication
- authorization
- rate limits
- versioning
- documentation
- abuse prevention
- privacy
- licensing
- provenance

Do NOT expose private or restricted information.

Do NOT launch a public API merely because it is technically possible.

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PART 17 – DATA LICENSING

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Document:

- public data rights
- source restrictions
- attribution requirements
- contributor rights
- third-party restrictions
- API usage terms

Do not assume all historical sources are freely redistributable.

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PART 18 – ADVANCED EXPORT

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Where legally and technically appropriate, support exports such as:

- research reports
- structured records
- citations
- source references
- selected collections

Respect licensing and privacy restrictions.

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PART 19 – PERFORMANCE AT SCALE

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Test safely using non-production environments.

Evaluate:

- larger datasets
- advanced search
- maps
- relationship graphs
- API workloads
- Android performance

Never perform uncontrolled production stress testing.

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PART 20 – SECURITY AT SCALE

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Review:

- new API surfaces
- research sharing
- collaboration permissions
- export permissions
- AI pipelines
- import pipelines

- public endpoints

Apply least privilege.

PART 21 – PRIVACY

Review privacy implications of:

- saved research
- collaboration
- contributor profiles
- research notes
- exports
- APIs
- AI processing

Collect only what is necessary.

PART 22 – GOVERNANCE

Create rules for:

- AI-generated suggestions
- historical corrections
- community research
- collaborative collections
- imported datasets
- public API access
- licensing

PART 23 – FEATURE GOVERNANCE

For every Phase 10 feature record:

- user problem
- evidence
- benefit
- complexity
- security impact
- privacy impact
- cost
- maintenance impact
- compatibility impact

Do not build speculative features without evidence.

PART 24 – DOCUMENTATION

Create/update as applicable:

docs/ADVANCED-RESEARCH.md
docs/ADVANCED-MAPS.md
docs/DATA-INTEROPERABILITY.md
docs/GLOBAL-EXPANSION.md
docs/AI-RESEARCH.md
docs/AI-REVIEW.md
docs/PUBLIC-API.md
docs/DATA-LICENSING.md
docs/ADVANCED-EXPORT.md
docs/LONG-TERM-ARCHIVE.md
docs/PHASE-10-ROADMAP.md

Document actual implementation only.

PHASE 10 ACCEPTANCE GATE

- [] Phase 9 baseline reviewed
- [] Advanced cemetery discovery
- [] Historical research
- [] Timeline exploration
- [] Relationship handling
- [] Advanced maps
- [] Research workspace
- [] Data interoperability
- [] Global expansion readiness
- [] Internationalization
- [] Community research
- [] Long-term archival architecture
- [] AI research assistance

- [] AI human-review controls
- [] Advanced search
- [] Public API evaluation
- [] Data licensing
- [] Advanced export
- [] Performance at scale
- [] Security at scale
- [] Privacy review
- [] Governance
- [] Feature governance
- [] Documentation

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FINAL REPORT

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PHASE 10 STATUS: READY / NOT READY

For every implemented capability report:

- what was built
- evidence
- tests performed
- limitations
- security considerations
- privacy considerations
- provenance considerations

Report actual metrics only.

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BLOCKERS

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List every issue preventing Phase 10 from being READY.

For each:

- severity
- component
- issue
- evidence
- impact
- remediation
- status

Never include secrets.

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ROADMAP

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NOW:
[items]

NEXT:
[items]

LATER:
[items]

DO NOT BUILD YET:
[items]

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STOP CONDITION

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Do NOT start Phase 11.

Do NOT add paid services without explicit approval.

Do NOT expose secrets.

Do NOT fabricate historical information.

Do NOT silently modify authoritative data.

Do NOT perform destructive bulk operations.

Do NOT perform uncontrolled production load testing.

Do NOT launch a public API without completing security, privacy, licensing and governance review.

Do NOT make AI-generated historical claims authoritative without human review.

STOP after the Phase 10 final report.

WAIT FOR EXPLICIT INSTRUCTIONS FOR PHASE 11.